

QUESTION 1 (15 Marks)**Marks**

- (a) Calculate the exact value of $\cos\left(\frac{5\pi}{12}\right)$. **3**
- (b) Evaluate the expression $x^2 + \frac{1}{x^2}$ when $x = \frac{1+2\sqrt{3}}{1-2\sqrt{3}}$. **3**
- (c) The second derivative of the function $y = f(x)$ is $\frac{d^2y}{dx^2} = \frac{1}{x^3} - \frac{2}{x^4}$. **2**

Show that the graph of $y = f(x)$ has a point of inflexion.

- (d) The point $P(x, y)$ moves so that its distance PA from the point $A(1, 5)$ is always twice its distance PB from the point $B(4, -1)$.
- (i) Show that the equation of the locus of P is $x^2 + y^2 - 10x + 6y + 14 = 0$. **2**
- (ii) Describe the locus. **2**
- (e) Sketch on a number plane the solution set for: **3**

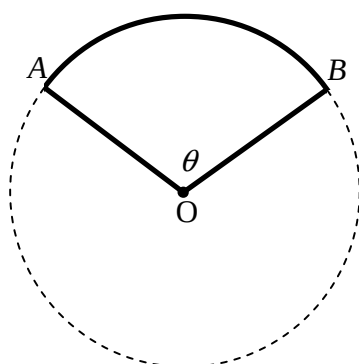
$$\{(x, y) : xy > 4\} \cap \{(x, y) : x - y \geq 0\}.$$

QUESTION 2 (15 Marks) START A NEW PAGE

- (a) Differentiate with respect to x :
- (i) $y = xe^x$ **1**
- (ii) $y = \log_x 2$ **2**
- (b) Solve the equation for x and verify the solution(s). **2**

$$\frac{|x-3|}{3-x} = x$$

- (c) A pond is designed in the shape of a sector of a circle, so that the perimeter of the pond is 200 m and the area is 1600 m^2 .



NOT TO SCALE

Calculate the value of the angle, θ radians, subtended at the centre of the circle. **4**

Question 2 continued over page

Question 2 continued

Marks

- (d) The point $P(-7, y)$ divides the interval AB in the ratio $a : b$, where A is $(-3, 1)$ and B is $(2, -9)$. Both a and b are integers.
- (i) Find the ratio $a : b$ expressed in simplest terms. 2
- (ii) Calculate the value of y . 1
- (e) The line m is the tangent to the curve $y = x^4 - 2x^3 - 5x$ at the point $P(2, -10)$. 3
Find the angle that m makes with the positive direction of the x -axis.
Give your answer to the nearest degree.

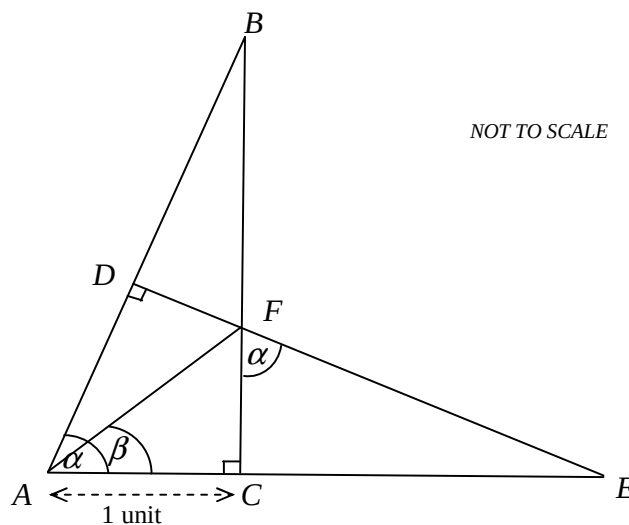
QUESTION 3 (15 Marks) START A NEW PAGE

- (a) State whether the function 2

$$f(x) = \frac{2^x - 1}{2^x + 1}$$

is **odd**, **even** or **neither odd nor even**, giving reasons.

- (b) In the diagram below $\triangle ABC$ is right-angled at C , with $\angle BAC = \alpha^\circ$. ED is drawn perpendicular to AB meeting BC at F such that $\angle EFC$ is also equal to α° . $\angle FAC = \beta^\circ$ and $AC = 1$ unit.



- (i) Copy the diagram and show that $BF = \tan \alpha - \tan \beta$ 1
- (ii) Express the length AE in terms of $\tan \alpha$ and $\tan \beta$. 1
- (iii) Prove that $\triangle BDF$ is similar to $\triangle ADE$. 2
- (iv) Hence prove the formula: 2

$$\tan(\alpha - \beta) = \frac{\tan \alpha - \tan \beta}{1 + \tan \alpha \tan \beta}.$$

Question 3 continued over page

Question 3 continued**Marks**

- (c) Solve the equation for x :

$$f(x) = f[f(x)] \text{ where } f(x) = x^2 - 2x.$$

3

- (d) If m and n are the roots of the equation $x^2 - x + 3 = 0$,
Find:

(i) the values of $m + n$, mn and $m^2 + n^2$.

2

(ii) the quadratic equation which has roots

2

$$\alpha = m^2 - mn \text{ and } \beta = n^2 - mn.$$

QUESTION 4 (15 Marks) START A NEW PAGE

- (a) Solve the equation for x :

3

$$9^x - 3^{x+2} = 10.$$

Give your answer to 2 decimal places.

- (b) Use first principles to find the derivative of $y = 8^x$.

3

You may assume that:

$$\lim_{u \rightarrow 0} \left[\frac{2^u - 1}{u} \right] = \ln 2.$$

- (c) Consider the function $y = \frac{12x^2}{x^3 + 4}$.

(i) Find $\frac{dy}{dx}$ in simplest terms.

2

(ii) Find the coordinates of all the stationary points for the function.

2

(iii) Determine the nature of the stationary points using the second derivative.

2

You may assume that $\frac{d^2y}{dx^2} = \frac{24(x^6 - 28x^3 + 16)}{(x^3 + 4)^3}$. (Do not show this)

(iv) Sketch the graph of $y = \frac{12x^2}{x^3 + 4}$, showing all turning points and asymptotes.

3**END OF EXAMINATION**

Question 1.

a) $\cos\left(\frac{5\pi}{12}\right) = \cos(75^\circ)$

$$= \cos(30^\circ + 45^\circ) =$$

$$= \cos 30^\circ \cos 45^\circ - \sin 30^\circ \sin 45^\circ$$

$$= \frac{\sqrt{3}}{2} \times \frac{1}{\sqrt{2}} - \frac{1}{2} \times \frac{1}{\sqrt{2}}$$

$$= \frac{\sqrt{3} - 1}{2\sqrt{2}}$$

① Correct Expansion
① + ① for correct values

b) $\left(\frac{1+2\sqrt{3}}{1-2\sqrt{3}}\right)^2 + \left(\frac{1-2\sqrt{3}}{1+2\sqrt{3}}\right)^2$

$$= \frac{1 + 4\sqrt{3} + 12}{1 - 4\sqrt{3} + 12} + \frac{1 - 4\sqrt{3} + 12}{1 + 4\sqrt{3} + 12}$$

$$= \frac{13 + 4\sqrt{3}}{13 - 4\sqrt{3}} + \frac{13 - 4\sqrt{3}}{13 + 4\sqrt{3}}$$

$$\frac{(13 + 4\sqrt{3})^2 + (13 - 4\sqrt{3})^2}{(13)^2 - (4\sqrt{3})^2}$$

$$= \frac{169 + 104\sqrt{3} + 48 + 169 - 104\sqrt{3} + 48}{169 - 48}$$

$$= \frac{434}{121}$$

① Expand

① Rationalize

① Simplify

c) $\frac{d^2y}{dx^2} = \frac{1}{x^3} - \frac{2}{x^4}$

For a possible point of inflexion $\frac{d^2y}{dx^2} = 0$.

$$\frac{x-2}{x^4} = 0$$

$$\therefore x = 2.$$

TEST concavity

x	1	2	3
$\frac{d^2y}{dx^2}$	-1	0	$+\frac{1}{81}$

$\therefore$ Change in concavity and $\frac{d^2y}{dx^2} = 0$
Point of inflexion when $x = 2$.

① solve $\frac{d^2y}{dx^2} = 0$

① Concavity Test

Question 1.

d) i) $PA = 2PB$
 $PA^2 = 4PB^2$

$$(x-1)^2 + (y-5)^2 = 4[(x-4)^2 + (y+1)^2]$$

$$x^2 - 2x + 1 + y^2 - 10y + 25 = 4(x^2 - 8x + 16 + y^2 + 2y + 1)$$

$$x^2 - 2x + 26 + y^2 - 10y = 4x^2 - 32x + 64 + 4y^2 + 8y + 4$$

$$0 = 3x^2 - 30x + 3y^2 + 18y + 42$$

$$0 = x^2 - 10x + y^2 + 6y + 14$$

ii) ~~$x^2 - 10x + 25 +$~~
 $y^2 + 6y + 9 = -14 + 25 + 9$

$$(x-5)^2 + (y+3)^2 = 20$$

Locus - circle, centre (5, -3)
 radius $2\sqrt{5}$
 units

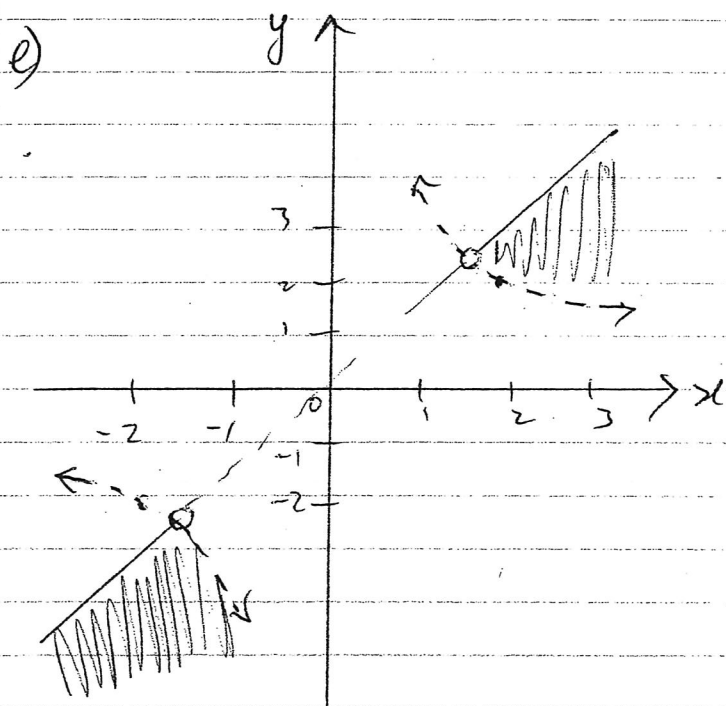
① distance equation

① Expansion and simplification

① Complete Squares + Factor

② circle

② centre + radius



① +

① each graph including boundary

① coloured region

① if circles at point of intersection not shown

Q2 a) $y' = e^x(x+1)$ (1m)

2m ii) $y = \frac{\ln 2}{\ln x}$ (1m) $y' = \frac{-\ln 2}{x(\ln x)^2}$ (1m)

1m b) Solve for x (1m)

Reject $x = -1, 3$ and verify $x = 1$

$\frac{|1-3|}{3-1} = \frac{|1-2|}{2} = \frac{2}{2} = 1$ (1m)

4m c) $\left. \begin{aligned} 2r + r\theta &= 200 \\ \frac{1}{2}r^2\theta &= 1600 \end{aligned} \right\}$ (1m)

$\left(\frac{200}{2+\theta}\right)^2 \theta = 3200$ (1m)

or $3200 = \left(\frac{200-2r}{r}\right)r^2$ (1m)

$\theta = 8$ or $\frac{1}{2}$ (1m)

Reject $\theta = 8$ as $\theta < 2\pi$ (1m)
 $\therefore \theta = \frac{1}{2}$ only

2m di) $\frac{2a-3b}{a+b} > -7$ (1m)

$a > b = -4 = 9$ (1m)

1m ii) $y = 9$ (1m)

3m e) $y' = 4x^3 - 6x^2 - 5$ (1m)

At $x=2$ $y' = 3$ (1m)

$\tan \theta = 3$ $\theta = 72^\circ$ (nearest degree) (1m)

$\ln x^2 \neq (\ln x)^2$

$y = \frac{\log_{10} 2}{\log_{10} x}$ 0m

most students got 1m as they forgot to verify

many drew graphs but did not verify $x = 1$

many forgot their formulae and can't proceed

most students forgot to reject $\theta = 8$

accept $\frac{a}{b} = -\frac{4}{9}$

$9a = -4b$ only 1/2m

many students write $y' = 4x^3 - 6x - 5$

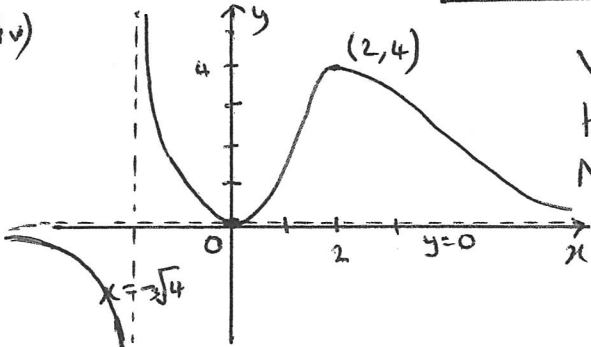
$\theta = 71.34^\circ$ 1/2m only

	Solution	Marks	Comments
(a)	$f(x) = \frac{2^x - 1}{2^x + 1}$ $f(-x) = \frac{2^{-x} - 1}{2^{-x} + 1}$ $= \frac{2^{-x} - 1}{2^{-x} + 1} \times \frac{2^x}{2^x}$ $= \frac{1 - 2^x}{1 + 2^x}$ $= -\left(\frac{2^x - 1}{2^x + 1}\right)$ $= -f(x)$ $f(-x) = \frac{2^{-x} - 1}{2^{-x} + 1} \times \frac{2^x}{2^x}$ $f(-x) = \frac{1 - 2^x}{1 + 2^x}$ $f(-x) = -\left(\frac{2^x - 1}{2^x + 1}\right)$ $f(-x) = -f(x)$ $\therefore f(x) \text{ is odd}$	2	$f(-x) = \frac{2^{-x} - 1}{2^{-x} + 1} \dots\dots\dots \frac{1}{2} \text{ mark}$ $f(-x) = \frac{1 - 2^x}{1 + 2^x} \dots\dots\dots \frac{1}{2} \text{ mark}$ $f(-x) = -\left(\frac{2^x - 1}{2^x + 1}\right) \dots\dots\dots \frac{1}{2} \text{ mark}$ $f(-x) = -f(x) \dots\dots\dots \frac{1}{2} \text{ mark}$ $\therefore f(x) \text{ is odd}$
(b)(i)	$BF = BC - FC$ $\frac{BC}{AC} = \tan \alpha$ $BC = AC \tan \alpha$ $= \tan \alpha$ $\frac{FC}{AC} = \tan \beta$ $FC = AC \tan \beta$ $= \tan \beta$ $BF = \tan \alpha - \tan \beta$	1	$BC = \tan \alpha \dots\dots\dots \frac{1}{2} \text{ mark}$ $FC = \tan \beta \dots\dots\dots \frac{1}{2} \text{ mark}$

(b)(ii)	$AE = AC + CE$ $\frac{CE}{CF} = \tan \alpha$ $CE = CF \tan \alpha$ $= \tan \alpha \tan \beta$ $AE = 1 + \tan \alpha \tan \beta$	1	$CE = \tan \alpha \tan \beta$½ mark $AE = 1 + \tan \alpha \tan \beta$½ mark
(b)(iii)	$\angle BFD = \alpha$ (vertically opposite angles are equal) In $\triangle BFD$ and $\triangle ADE$ $\angle BDE = \angle ADE$ (both 90°) $\angle BFD = \angle DAE$ (both $= \alpha$) $\triangle BFD \parallel \triangle ADE$ (equiangular) Could also have $\angle BDF = \angle AED$ ($= 90^\circ - \alpha$) (angle sum of triangles $= 180^\circ$)	2	$\angle BFD = \alpha$½ mark 2 equal angles½ mark each angle Test (equiangular).....½ mark
(b)(iv)	$\frac{DF}{DA} = \tan(\alpha - \beta)$ $\frac{BF}{AE} = \frac{DF}{DA}$ (ratios of corresponding sides in similar triangles are equal) $\tan(\alpha - \beta) = \frac{\tan \alpha - \tan \beta}{1 + \tan \alpha \tan \beta}$	2	$\frac{DF}{DA} = \tan(\alpha - \beta)$ 1 mark $\frac{BF}{AE} = \frac{DF}{DA}$ (ratios of corresponding sides in similar triangles are equal)1 mark

(c)	$x^2 - 2x = (x^2 - 2x)^2 - 2(x^2 - 2x)$ $\text{let } u = x^2 - 2x$ $u = u^2 - 2u$ $u^2 - 3u = 0$ $u(u - 3) = 0$ $u = 0 \text{ or } 3$ $x^2 - 2x = 0 \text{ or } x^2 - 2x = 3$ $x(x - 2) = 0 \text{ or } x^2 - 2x - 3 = 0$ $x = 0 \text{ or } 2 \text{ or } (x - 3)(x + 1) = 0$ $x = 3 \text{ or } -1$ $\therefore x = 0 \text{ or } 2 \text{ or } 3 \text{ or } -1$ * can also expand, simplify and use remainder theorem to get roots $x^4 - 4x^3 + x^2 + 6x = 0$ $x(x - 2)(x - 3)(x + 1) = 0$	3	$x^2 - 2x = (x^2 - 2x)^2 - 2(x^2 - 2x)$ Correct equation 1 mark $\frac{1}{2}$ mark for each solution 0 mark for correct solutions from incorrect working ** $(x^2 - 2x)^2 - 2(x^2 - 2x) = 0$ $\Rightarrow x(x - 2) = 0$ $\Rightarrow x = 0 \text{ or } 2$ max. 1 mark
(d)(i)	$m + n = 1$ $mn = 3$ $m^2 + n^2 = (m + n)^2 - 2mn$ $= 1 - 6$ $= -5$	2	$m + n = 1$ $\frac{1}{2}$ mark $mn = 3$ $\frac{1}{2}$ mark $m^2 + n^2 = (m + n)^2 - 2mn$ $= 1 - 6$ 1 mark $= -5$
(d)(ii)	Quadratic equation is $x^2 - (\alpha + \beta)x + \alpha\beta = 0$ $\alpha + \beta = (m^2 - mn) + (n^2 - mn)$ $= m^2 + n^2 - 2mn$ $= (m + n)^2 - 4mn$ $= 1 - 12$ $= -11$ $\alpha\beta = (m^2 - mn)(n^2 - mn)$ $= m^2n^2 - m^3n - mn^3 + m^2n^2$ $= 2(mn)^2 - mn(m^2 + n^2)$ $= 2(9) - 3(-5)$ $= 33$ Quadratic equation is $x^2 + 11x + 33 = 0$		$\alpha + \beta = -11$ $\frac{1}{2}$ mark $\alpha\beta = 33$ $\frac{1}{2}$ mark +11x (or correct sign for their coeff. of x) $\frac{1}{2}$ mark correct position of (their) x-coefficient and constant and = 0 $\frac{1}{2}$ mark

J:\Maths\Suggested Mk solns template_V2.doc

Suggested Solutions	Marks	Marker's Comments
a) $3^{2x} - 9 \cdot 3^x - 10 = 0$ $u^2 - 9u - 10 = 0$ (Let $u = 3^x$) $(u-10)(u+1) = 0$ $u = 10$ or -1 But $3^x > 0 \therefore 3^x = 10$ only $x = \frac{\log 10}{\log 3}$ <u>$= 2.10$ (2 PP)</u>	1 1 1	Too many people took logs of $3^{2x} - 3^{x+2}$ and got no marks. Four people arrived at 2.10 with no valid working - no marks.
b) $\frac{dy}{dx} = \lim_{h \rightarrow 0} \left\{ \frac{2^{3(x+h)} - 2^{3x}}{h} \right\} = 2^{3x} \lim_{h \rightarrow 0} \left\{ \frac{2^{3h} - 1}{h} \right\}$ $= 2^{3x} \lim_{h \rightarrow 0} \left\{ \left(\frac{2^{3h} - 1}{3h} \right) 3 \right\}$ $= 2^{3x} \lim_{h \rightarrow 0} \left(\frac{2^{3h} - 1}{3h} \right) \cdot 3$ <u>$= 2^{3x} \cdot 3 \cdot \ln 2 = 8^x \ln 8$</u>	1 1 1	One mark for 1st principle definition. One mark for knowing answer. Some justification required that $\lim_{h \rightarrow 0} \frac{8^h - 1}{h} = \ln 8$
c) i) $\frac{dy}{dx} = \frac{(x^3 + 4)24x - 12x^2(3x^2)}{(x^3 + 4)^2} = \underline{\underline{\frac{-12x^4 + 96x}{(x^3 + 4)^2}}}$ ii) At S.P., $\frac{dy}{dx} = 0 \Rightarrow x(2-x)(x^2 + 2x + 4) = 0$ $\therefore x = 0$ or $x = 2 \Rightarrow$ S.P. at <u>$(0,0)$ and $(2,4)$</u> iii) At $(0,0)$, $\frac{d^2y}{dx^2} = 6 \therefore$ Concave up <u>Local Minimum at $(0,0)$</u> at $(2,4)$, $\frac{d^2y}{dx^2} = -2 \therefore$ Concave down <u>Local Maximum at $(2,4)$</u> iv)  Vert Asym $x = -\sqrt[3]{4}$ Horiz Asym $y = 0$ No intercepts except $(0,0)$	2 2 1 1 1/2 per asym 1/2 per SP 1 for shape and axes.	$\frac{12x(8-x)}{(x^3+4)^2}$ better Too many numerical errors in Quotient Rule. S.P. are points not x values. In part (ii) some mention of concavity was required. Scales were scarier than I expected. At least 0 and 1 other value on each axis.